

Survival Analysis for Metastatic Breast Cancer Patients

Introduction

Breast cancer is the most commonly diagnosed cancer in women, and there are more than 3.1 million with a history of breast cancer as of January 2018[1]. In year 2018, there are 266,120 estimated new cases and an expected mortality of 40,920[1]. It is the second leading cause of cancer death among women [1]. The diagnosis of metastatic cancer (stage IV cancer), cancer that has spread to other parts of the body, is particularly lethal, with a 5 year survival rate as low as 22 percent [1]. Yet, although metastatic cancer is incurable, it is treatable. Treatments such as chemotherapy can slow the growth and spread of cancer, and palliative care can relieve the symptoms from cancer or the side effects of treatments to improve quality of life.

Nonetheless, there still exists barriers to access of palliative care in many parts of the world, mostly because of the cure-versus-care mentality prevalent in the medical community. For instance, ministries of health often place a higher value on interventions that could cure diseases than those that can improve the quality of life, which results in limited access to pain relief for many patients. Considering the significance of palliative care for the well-being of surviving metastatic cancer patients, it would be important to develop policies that could better address the rising need of palliative care.

Below is a list of definitions for the major terms in our project:

- 1) Long term surviving patients: Patients who have survived for more than 5 years after diagnosis
- 2) 5 year survival rate: A statistic that describes the percentage of people who will be alive 5 years after cancer diagnosis. It includes those in remission (temporary or permanent absence of disease) and those still receiving treatment. It does not include those who die from other diseases.
- 3) Metastatic cancer (stage IV cancer): Cancer that has spread to other parts of the body, most commonly the liver, lungs, lymph nodes, and bones. Breast cancer that has spread to the lungs is still called “breast cancer”. Thus, the cause of death for patients diagnosed with breast cancer may not necessarily be the cancer in the breasts.
- 4) Palliative care; A non-curative care for patients with life threatening diseases (cancer, heart disease, dementia, HIV/AIDS, ALS, Parkinson’s disease) that prevents or treats the symptoms and side effects of the disease and its treatment. It can be given alongside treatments, and addresses the patients’ psychological, social, and spiritual problems, thereby providing care for the patient as a whole and not just the disease.

Objective

Our main objective is to assess whether there is a clear trend in survival time for metastatic breast cancer patients diagnosed between 1988 and 2009--whether survival time has

increased or decreased over the years. Additionally we want to determine if this survival time is affected by a patient's age at diagnosis, race or marital status, and if so, how it is affected. We will be analyzing a SEER database to make descriptive inferences about the data on breast cancer patients. Our specific analytic goals are expressed in the following questions:

- 1) Regarding the trend in survival time for breast cancer patients:
 - a) Are the Kaplan-Meier survival curves, stratified by year of diagnosis, significantly different from each other?
 - b) Does an increase in year of diagnosis correspond to a lower hazard ratio for death?

(a) is the analytic goal for the Kaplan-Meier method, (b) is the analytic goal for the Cox proportional hazards regression analysis method.
- 2) Regarding the potential effect of a patient's age at diagnosis, race or marital status on that patient's survival time:
 - a) Are the Kaplan-Meier survival curves, stratified by age at diagnosis and race respectively, significantly different from each other?
 - b) Does an increase in age at diagnosis correspond to an increased hazard ratio for death?
 - c) Do patients of different races have different hazard ratios for death?
 - d) Do patients with different marital statuses have different hazard ratios for death?

(a) is the analytic goal for the Kaplan-Meier method; (b), (c) and (d) are analytic goals for the Cox proportional hazards regression analysis method.

If our analysis does show that there has been an increase in survival rate over time, it would place greater emphasis on the need for the improvement of policies on palliative care.

Data (Study Population)

SEER (Surveillance, Epidemiology, and End Results) is a U.S. cancer registry database collected years 1973 to 2014 from different hospitals in participating states (California, Connecticut, Detroit, Hawaii, Iowa, New Mexico, Seattle, Utah, Atlanta, Alaska, Georgia, Kentucky, Louisiana, New Jersey), accounting for approximately 34.6% of the US population. The rows are patient ID, and the columns are different factors such as age, race/ethnicity, year of diagnosis, ER/PR status, marriage status. The distribution is approximately normal and the sample space is $n=1631572$. No conditioning is performed on modeling. The patient data is randomized, but there could be a correlation structure within different states.

Model

1. The statistical model for Kaplan Meier is $\widehat{\text{Var}}(\widehat{S}(t)) = \widehat{S}(t)^2 \sum_{i: t_i \leq t} \frac{d_i}{n_i(n_i - d_i)}$, where d_i is the number of cases and n_i is the total number of observations, for $t_i < t$. In this case, d_i is the

number of metastatic breast cancer patients who were diagnosed between years 1988-2009, and n_i is the number of patients diagnosed between years 1988-2014 who survived more than five years.

- a. $S(t) = P\{T > t\} = 1 - F(t)$ The survival function, $S(t)$ gives the probability of surviving beyond time t . Because $S(t)$ is a probability, it is positive and ranges from 0 to 1. It is defined as $S(0) = 1$ and as t approaches ∞ , $S(t)$ approaches 0. The Kaplan-Meier estimator incorporates information from all of the observations available, both censored and uncensored, by considering any point in time as a series of steps defined by the observed survival and censored times. The survival curve describes the relationship between the probability of survival and time. [2]
 - b. Kaplan Meier was chosen as the method for analyzing survival curves, because it incorporates the censored data in our study that exists due to the nature of survival analysis. If the variable of interest was whether the studied patients survived more than five years or not, and there were no censored patients, simple logistic regression could have been used with the outcome as [more than five years of survival or not] and the years of diagnosis as the predictors.
2. The statistical model for Cox Proportional Hazards Regression
- a. $h(t) = P\{t < T < (t + \Delta) \mid T > t\} = f(t) / (1 - F(t)) = f(t) / S(t)$ The hazard function describes the concept of the risk of death in an interval after time t , conditional on the subject having survived to time t . It is the probability that an individual dies somewhere between t and $(t + \Delta)$, divided by the probability that the individual survived beyond time t . It attempts to quantify the instantaneous risk that an event will take place at time t given that the subject survived to time t . [2]
 - b. A multiple logistic regression could have been used with the factors (age at diagnosis, race, ER/PR receptivity, etc) with the predictor and the [more than five years of survival or not] as the outcome. However, due to the censoring nature of survival data, Cox regression analysis was used instead.

Methods

This retrospective cohort analysis used the SEER data registry (diagnosed 1988-2014), patients first diagnosed with metastatic (stage IV) breast cancer whose length of survival till death have been recorded were identified. Patients were divided into age group 1 (49 years and under), age group 2 (50-59), age group 3 (60-69), age group 4 (70-79), and age group 5 (80 and above). The different race and ethnic groups were also categorized as Caucasian, Hispanic (Mexican, Puerto Rican, Cuban, Spanish/Hispanic/Latino, Dominican), Black, Asian, and Other. Marriage status was categorized as single (never married, separated, divorced, or widowed), married, and unmarried (living with a partner). ER/PR receptivity was categorized as; negative (if at least either of ER or PR is negative), positive (if ER is positive or borderline, and PR is positive or borderline), and unknown (if ER and PR are both unknown). For all variables, data

with unentered variables were excluded for Kaplan Meier and Cox Regression analyses. The following diagram describes how we restricted the dataset to include only patients of interest for our analyses.

Patient demographics

Between 1973-2014, there were 1631572 breast cancer patients reported in the SEER database. Between the years 1988-2009 that were used for this study, there were 35649 patients that were first diagnosed with metastatic breast cancer (AJCC stage IV), and had a length of survival months recorded. The characteristics of these patients are summarized in Table 1.

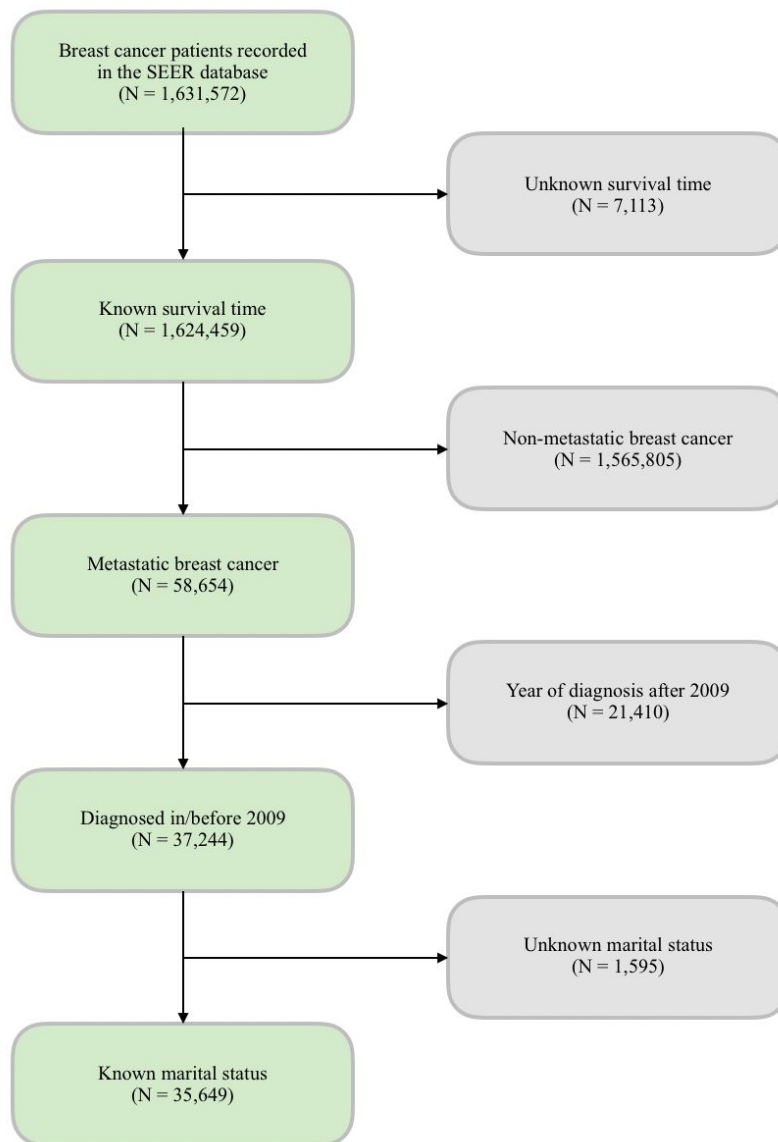


Figure 1. Flow diagram for the selection of the study cohort used in regression analysis.

To consider the potential bias on survival analysis caused by censored data, Kaplan Meier curves were used to compare survival rates across different groups. In analyzing the effect of predictor variables, Cox regression analysis was used.

For the Kaplan Meier analysis, the years of diagnosis were divided into group 1 (1988-1993), group 2 (1993-1998), group 3 (1998-2003), group 4 (2003-2008), and group 5 (2008-2013).

To compare the survival rate between years of diagnosis (1988-2013), Kaplan-Meier curves were generated and the log rank test was applied to analyze the difference among the curves. The same method was applied to compare survival rate between different age groups across the five groups of years of diagnosis (1988-2013).

Results

Kaplan Meier Curves, Log-ranked for significance

- I. Kaplan Meier curves were used to analyze difference in survival over the different groups of years of diagnosis; group 1 (1988-1993), group 2 (1993-1998), group 3 (1998-2003), group 4 (2003-2009). Although the rank tests for homogeneity for groups 1 and 2 did not show significant difference (1988-1998, $\chi^2=30.068$, $P=0.813$), groups 2 and 3 groups 3 and 4 showed significant difference. Groups 2 and 3, years 1998-2003 log-ranking showed $\chi^2=74.564$, $P<.0001$, and groups 3 and 4, years 2003-2009, were $\chi^2=74.564$, $P<.0001$. There was significant improvement in survival overall between the years 1988-2009 ($\chi^2=105.843$, $P<.0001$).
- II. Next, survival curves amongst age groups were compared by years of diagnosis. The age groups are, group 1 (49 years and under), age group 2 (50-59), age group 3 (60-69), age group 4 (70-79), and age group 5 (80 and above). The years of diagnosis are; group 1 (1988-1993), group 2 (1993-1998), group 3 (1998-2003), group 4 (2003-2009).
 - A. The Kaplan Meier curves for metastatic breast cancer patients diagnosed between years 1988-1993, for different age groups. ($\chi^2=57.898$, $P<.0001$).

<i>Covariance Matrix for the Log-Rank Statistics of patients diagnosed between years 1988-1993</i>					
<i>agegrp</i>	<i>1</i>	<i>2</i>	<i>3</i>	<i>4</i>	<i>5</i>
<i>1</i>	<i>809.699</i>	<i>-237.599</i>	<i>-252/364</i>	<i>-210.434</i>	<i>-109.303</i>
<i>2</i>	<i>0237.599</i>	<i>750.690</i>	<i>-224.902</i>	<i>-189.126</i>	<i>-99.063</i>

3	0252.364	-224.902	788.145	-203.496	-107.384
4	-210.434	-189.126	-203.496	696.756	-93.701
5	-109.303	-99.063	-107.384	-93.701	409.452

B. The Kaplan Meier curves for metastatic breast cancer patients diagnosed between years 1993-1997, for different age groups. $\chi^2=213.327$, $P<.0001$.

<i>Covariance Matrix for the Log-Rank Statistics of patients diagnosed between years 1993-1998</i>					
<i>agegrp</i>	<i>1</i>	<i>2</i>	<i>3</i>	<i>4</i>	<i>5</i>
<i>1</i>	345.269	-81.845	-117.140	-97.082	-49.203
<i>2</i>	-81.845	322.131	-106.498	-88.676	-45.113
<i>3</i>	-117.140	-106.498	422.252	-131.462	-67.152
<i>4</i>	097.082	-88.676	0131.462	374.662	-57.442
<i>5</i>	-49.203	-45.113	-67.152	-57.442	218.910

C. The Kaplan Meier curves for metastatic breast cancer patients diagnosed between years 1998-2003, for different age groups. $\chi^2=565.355$, $P<.0001$.

<i>Covariance Matrix for the Log-Rank Statistics of patients diagnosed between years 1998-2003</i>					
<i>agegrp</i>	<i>1</i>	<i>2</i>	<i>3</i>	<i>4</i>	<i>5</i>
<i>1</i>	1392.43	-478.75	-408.51	-340.23	-169.94
<i>2</i>	-478.75	1498.45	-453.01	-380.17	-186.51
<i>3</i>	-408.51	-106.498	422.252	-131.462	-67.152
<i>4</i>	097.082	-88.676	0131.462	374.662	-57.442
<i>5</i>	-49.203	-45.113	-67.152	-57.442	218.910

D. The Kaplan Meier curves for metastatic breast cancer patients diagnosed between years 2003-2009, for different age groups. $\chi^2=1091.004$, $P<.0001$.

<i>Covariance Matrix for the Log-Rank Statistics of patients diagnosed between years 2003-2009</i>					
<i>agegrp</i>	<i>1</i>	<i>2</i>	<i>3</i>	<i>4</i>	<i>5</i>
<i>1</i>	2495.84	-856.33	-801.19	-535.05	-303.27
<i>2</i>	-856.33	2562.38	-826.77	-557.17	-322.10
<i>3</i>	-801.19	-826.77	2454.07	-522.69	-303.42
<i>4</i>	-535/05	-557.17	-522.69 _q	1826.76	-211.84
<i>5</i>	-303.27	-322.10	-303.42	-211.84	1140.63

The log rank statistic is used to compare the five different curves generated, which testing for the null hypothesis that the curves are exactly the same. Which means, that the probability of death occurring for a particular age group is the same for another. Since $P < .0001$ for all all years of diagnosis groups, survival seems to be dependant on the age of diagnosis for all years considered (1988-1993). Furthermore, the χ^2 value is increasing overtime, which shows that although all instances had significant P-value, because the degrees of freedom are the same, the disparity in survival amongst different age groups have increased over the years. Comparing the curves on the graph and studying the covariance matrix for the log-rank tells us that although all age groups have improved survival, the younger age groups seem to have improved more than the older age groups (4-5).

III. To confirm the above result, survival curves amongst year of diagnosis were compared by age groups, stratified by years of diagnosis. Again, the age groups are, group 1 (49 years and under), age group 2 (50-59), age group 3 (60-69), age group 4 (70-79), and age group 5 (89 and above). The years of diagnosis are; group 1 (1988-1993), group 2 (1993-1998), group 3 (1998-2003), group 4 (2003-2009).

- A. The Kaplan Meier curves for metastatic breast cancer patients diagnosed at age 49 years and under, had significantly different curves between years of diagnosis, with $\chi^2=98.853$, $P < .0001$.
- B. The Kaplan Meier curves for metastatic breast cancer patients diagnosed at ages between 50-59, had significantly different curves between years of diagnosis, with $\chi^2=47.745$, $P < .0001$.
- C. The Kaplan Meier curves for metastatic breast cancer patients diagnosed at ages between 60-69, had significantly different curves between years of diagnosis, with $\chi^2=57.700$, $P < .0001$.

D. The Kaplan Meier curves for metastatic breast cancer patients diagnosed at ages between 70-79, had significantly different curves between years of diagnosis, with $\chi^2=33.2667$, $P<.0001$.

E. The Kaplan Meier curves for metastatic breast cancer patients diagnosed at ages between 80 and above, had significantly different curves between years of diagnosis, with $\chi^2=9.6592$, $P<.0001$.

Although all groups experienced significant difference between curves, which meant that they all experienced improvement, the χ^2 values had increased with age. This suggests that the older age groups see less difference between survival curves over the years, compared to the younger age groups. This in turn means that the younger age groups have experienced higher improvement in survival over the years.

IV. Finally, survival curves were analyzed for metastatic breast cancer patients in different age groups (1-5), for different races. Races; Caucasian, Hispanic (Mexican, Puerto Rican, Cuban, Spanish/Hispanic/Latino, Dominican), Black, Asian, and Other.

A. The Kaplan Meier curves for metastatic breast cancer patients diagnosed between years 1988-2009, for the Asian population. $\chi^2= 75.508$, $P<.0001$.

B. The Kaplan Meier curves for metastatic breast cancer patients diagnosed between years 1988-2009, for the Black population. $\chi^2= 129.604$, $P<.0001$.

C. The Kaplan Meier curves for metastatic breast cancer patients diagnosed between years 1988-2009, for the Hispanic population. $\chi^2=71.947$, $P<.0001$.

D. The Kaplan Meier curves for metastatic breast cancer patients diagnosed between years 1988-2009, for the Caucasian population. $\chi^2= 1750.815$, $P<.0001$. Although the survival curves differed greatly in each race, suggesting that survival had improved in all race groups, the curves did not seem to vary between races.

Modeling the effect of the following predictors on the hazard ratio for death from metastatic breast cancer with Cox proportional hazards regression

I. Year of diagnosis

Analysis of Maximum Likelihood Estimates						
Parameter	DF	Parameter Estimate	Standard Error	Chi-Square	Pr > ChiSq	Hazard Ratio
year_numeric	1	-0.01708	0.0009467	325.4836	<.0001	0.983

The association between year of diagnosis and time to death appears to be statistically significant with a parameter estimate of -0.01708 ($p<0.0001$) relative to a one unit increase in year of

diagnosis. The hazards ratio computed by the Cox proportional hazards regression indicates that a one unit increase in year of diagnosis is associated with a 1.7% decrease in the expected hazard for death. In other words, patients diagnosed later have a lower hazard for death and therefore have longer survival times.

II. Age at diagnosis

Analysis of Maximum Likelihood Estimates						
Parameter	DF	Parameter Estimate	Standard Error	Chi-Square	Pr > ChiSq	Hazard Ratio
age_numeric	1	0.01733	0.0004009	1869.1845	<.0001	1.017

The association between age at diagnosis and time to death appears to be statistically significant with a parameter estimate of 0.01733 ($p < 0.0001$) relative to a one unit increase in age at diagnosis. The hazards ratio computed by the Cox proportional hazards regression indicates that a one unit increase in age at diagnosis is associated with a 1.7% increase in the expected hazard for death. In other words, patients diagnosed at an older age have a higher hazard for death and therefore have shorter survival times.

III. Marriage status

Analysis of Maximum Likelihood Estimates								
Parameter		DF	Parameter Estimate	Standard Error	Chi-Square	Pr > ChiSq	Hazard Ratio	Label
marriage_stat	marrie	1	-0.30644	0.01123	744.9856	<.0001	0.736	marriage_stat marrie

The association between marriage status and time to death appears to be statistically significant with a parameter estimate of -0.30644 ($p < 0.0001$). The hazards ratio computed by the Cox proportional hazards regression indicates that having a marriage status of “married” is associated with a 26.4% decrease in the expected hazard for death. In other words, patients who are married have a lower hazard for death and therefore have longer survival times.

IV. Race

Analysis of Maximum Likelihood Estimates								
Parameter		DF	Parameter Estimate	Standard Error	Chi-Square	Pr > ChiSq	Hazard Ratio	Label
race	A	1	-0.19165	0.02548	56.5944	<.0001	0.826	race A
race	B	1	0.19064	0.01543	152.6511	<.0001	1.210	race B
race	H	1	-0.08757	0.02245	15.2105	<.0001	0.916	race H
race	O	1	-0.06340	0.07566	0.7020	0.4021	0.939	race O

The association between race and time to death appears to be statistically significant for three race groups. The A race group (Asian) has a parameter estimate of -0.19165 ($p<0.0001$), the B race group (Black) has a parameter estimate of 0.19064 ($p<0.0001$), and the H race group (Hispanic) has a parameter estimate of -0.08757 ($p<0.0001$). The hazards ratios computed by the Cox proportional hazards regression indicate that being Asian is associated with a 17.4% decrease in the expected hazard for death, being Hispanic is associated with an 8.4% decrease in the expected hazard for death, while being black is associated with a 21% increase in the expected hazard for death.

V. Year of diagnosis and Age at diagnosis

Analysis of Maximum Likelihood Estimates							
Parameter	DF	Parameter Estimate	Standard Error	Chi-Square	Pr > ChiSq	Hazard Ratio	Label
year_numeric	1	-0.03668	0.00461	63.4045	<.0001	.	
age_numeric	1	-6.67402	1.50066	19.7792	<.0001	.	
year_num*age_numeri	1	0.00342	0.0007496	20.7659	<.0001	.	year_numeric * age_numeric

The association between the interaction of year of diagnosis and age at diagnosis and time to death appears to be statistically significant ($p<0.0001$).

VI. Year of diagnosis and Marriage status

Type 3 Tests			
Effect	DF	Wald Chi-Square	Pr > ChiSq
year_numeric	1	303.0872	<.0001
marriage_stat	1	4.2408	0.0395
year_num*marriage_s	1	4.5422	0.0331

Analysis of Maximum Likelihood Estimates								
Parameter		DF	Parameter Estimate	Standard Error	Chi-Square	Pr > ChiSq	Hazard Ratio	Label
year_numeric		1	-0.01656	0.00140	139.6425	<.0001	.	
marriage_stat	marrie	1	8.84307	4.29417	4.2408	0.0395	.	marriage_stat marrie
marriage_stat	single	0	0	.	.	.	.	marriage_stat single
year_nume*marriage_s	marrie	1	-0.00457	0.00215	4.5422	0.0331	.	marriage_stat marrie * year_numeric
year_nume*marriage_s	single	0	0	.	.	.	.	marriage_stat single * year_numeric

The association between the interaction of year of diagnosis and marriage status and time to death does not appear to be statistically significant (p=0.0331).

VII. Age at diagnosis and Race

Type 3 Tests			
Effect	DF	Wald Chi-Square	Pr > ChiSq
age_numeric	1	103.9482	<.0001
race	4	158.5285	<.0001
age_numeric*race	4	81.7698	<.0001

Analysis of Maximum Likelihood Estimates								
Parameter		DF	Parameter Estimate	Standard Error	Chi-Square	Pr > ChiSq	Hazard Ratio	Label
age_numeric		1	0.19640	0.00485	1639.7938	<.0001	.	
race	A	1	0.25503	0.10634	5.7516	0.0165	.	race A
race	B	1	0.77262	0.06354	147.8625	<.0001	.	race B
race	H	1	0.42185	0.08577	24.1907	<.0001	.	race H
race	O	1	0.64033	0.32706	3.8332	0.0502	.	race O
race	W	0	0	.	.	.	.	race W
age_numeric*race	A	1	-0.06144	0.01888	10.5859	0.0011	.	race A * age_numeric
age_numeric*race	B	1	-0.08624	0.01091	62.4734	<.0001	.	race B * age_numeric
age_numeric*race	H	1	-0.07006	0.01523	21.1533	<.0001	.	race H * age_numeric
age_numeric*race	O	1	-0.11085	0.05887	3.5457	0.0597	.	race O * age_numeric
age_numeric*race	W	0	0	.	.	.	.	race W * age_numeric

The association between the interaction of age at diagnosis and race and time to death appears to be statistically significant ($p < 0.0001$). The interaction between age at diagnosis and race is particularly strong for people of Black and Hispanic races.

VIII. Race and Marriage status

Type 3 Tests			
Effect	DF	Wald Chi-Square	Pr > ChiSq
race	4	165.6311	<.0001
marriage_stat	1	39.6509	<.0001
race*marriage_stat	4	18.2838	0.0011

Analysis of Maximum Likelihood Estimates									
Parameter			DF	Parameter Estimate	Standard Error	Chi-Square	Pr > ChiSq	Hazard Ratio	Label
race	A		1	-0.20225	0.03753	29.0426	<.0001	.	race A
race	B		1	0.12394	0.01900	42.5466	<.0001	.	race B
race	H		1	-0.15937	0.03085	26.6961	<.0001	.	race H
race	O		1	-0.16988	0.10300	2.7204	0.0991	.	race O
race	W		0	0	.	.	.	.	race W
marriage_stat	marrie		1	-0.31923	0.01346	562.5707	<.0001	.	marriage_stat marrie
marriage_stat	single		0	0	.	.	.	.	marriage_stat single
race*marriage_stat	A	marrie	1	0.06560	0.05181	1.6030	0.2055	.	race A * marriage_stat marrie
race*marriage_stat	A	single	0	0	.	.	.	.	race A * marriage_stat single
race*marriage_stat	B	marrie	1	0.07920	0.03426	5.3441	0.0208	.	race B * marriage_stat marrie
race*marriage_stat	B	single	0	0	.	.	.	.	race B * marriage_stat single
race*marriage_stat	H	marrie	1	0.16108	0.04552	12.5230	0.0004	.	race H * marriage_stat marrie
race*marriage_stat	H	single	0	0	.	.	.	.	race H * marriage_stat single
race*marriage_stat	O	marrie	1	0.20981	0.15506	1.8309	0.1760	.	race O * marriage_stat marrie
race*marriage_stat	O	single	0	0	.	.	.	.	race O * marriage_stat single
race*marriage_stat	W	marrie	0	0	.	.	.	.	race W * marriage_stat marrie
race*marriage_stat	W	single	0	0	.	.	.	.	race W * marriage_stat single

The association between the interaction of race and marriage status and time to death does not appear to be that statistically significant ($p = 0.0011$).

IX. Age and Age

Analysis of Maximum Likelihood Estimates							
Parameter	DF	Parameter Estimate	Standard Error	Chi-Square	Pr > ChiSq	Hazard Ratio	Label
age_numeric	1	-0.02431	0.00285	72.6741	<.0001	.	
age_numer*age_numeri	1	0.0003317	0.0000225	216.5699	<.0001	.	age_numeric * age_numeric

The association between the interaction between age and age and time to death does appear to be statistically significant ($p < 0.0001$).

X. Final model

Our final model models the association between the predictors year of diagnosis, age at diagnosis and marriage status with time to death, and includes interaction terms for age and age, and age and year of diagnosis. Race was not included in the model because the association between race and time to death was not statistically significant for all race groups when a Cox proportional hazards model was performed with race as the only predictor. Additionally, Table 1 shows that there are vastly different numbers of patients in each race group in our dataset, which introduces possible biases into an analyses that includes race.

Analysis of Maximum Likelihood Estimates								
Parameter		DF	Parameter Estimate	Standard Error	Chi-Square	Pr > ChiSq	Hazard Ratio	Label
year_numeric		1	-0.03722	0.00432	74.0939	<.0001	.	
age_numeric		1	-0.71743	0.13109	29.9496	<.0001	.	
marriage_stat	marrie	1	-0.20238	0.01162	303.5573	<.0001	0.817	marriage_stat marrie
age_numer*age_numeri		1	0.0002874	0.0000227	159.7036	<.0001	.	age_numeric * age_numeric
year_num*age_numeri		1	0.0003481	0.0000655	28.2507	<.0001	.	year_numeric * age_numeric

Conclusions

Our Kaplan Meier analysis showed that there was significant improvement of survival over the years, and that the rate of improvement differs significantly between age groups, but not among races or marriage statuses.

Overall, our analyses show that patients diagnosed with metastatic breast cancer more recently, generally have lower hazards for death (longer survival times). Additionally, survival time appears to be affected by the age and marital status of a patient; patients diagnosed at an older age experienced worse survival than those from younger age groups, and black patients experienced worse survival than Asian or Hispanic patients.

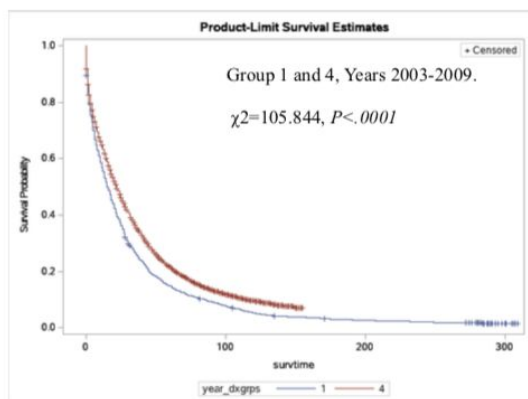
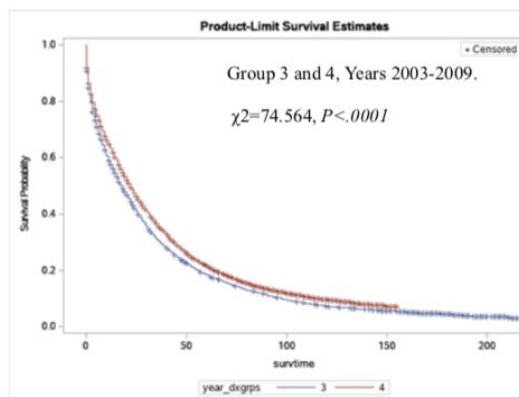
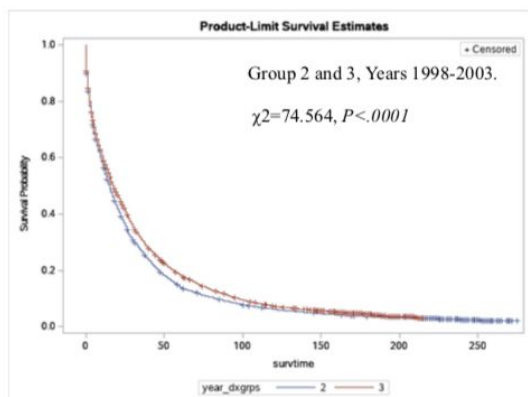
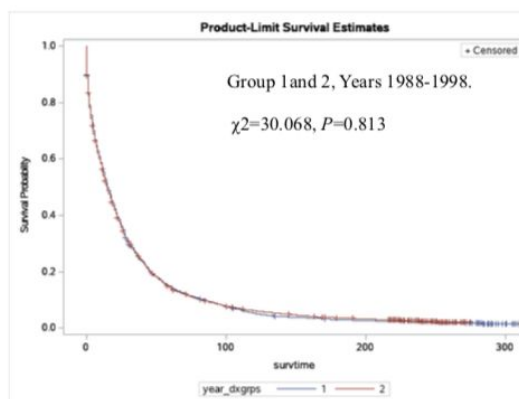
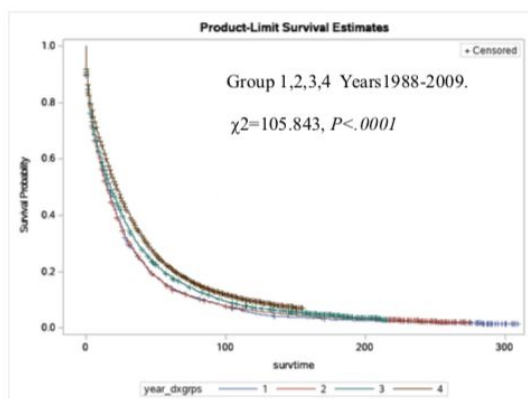
Figures

Table 1. Characteristics of breast cancer patients with AJCC stage IV from SEER Database from 1988-2013.

Characteristics	Number	%
Age		
49<	6959	19.43
50-59	7973	22.26
60-69	8232	22.98
70-79	7287	20.34
80>	5372	15.00
Year of diagnosis		
1988-1993	2367	6.61
1993-1998	4888	13.64
1998-2003	8954	25.00
2003-2008	13290	37.10
2008-2013	6324	17.65
Race		
Asian	1971	5.50
Black	5547	15.49
Hispanic	2554	7.13
White	25512	71.23
Other	233	0.65
Marriage status		
Married	15047	43.89
Single	19235	56.11
ER/PR status		
Positive	19498	54.43

Negative	7000	19.54
Unknown	9325	26.03

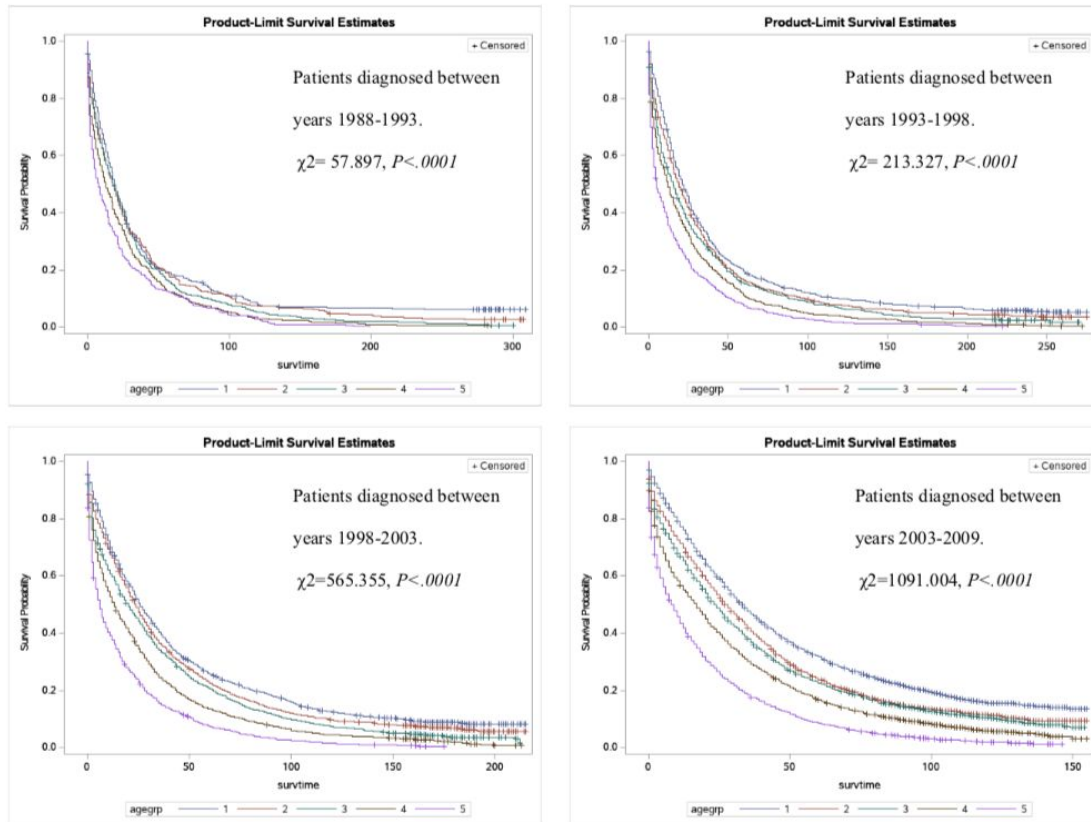
I. The Kaplan Meier curves for metastatic breast cancer patients with different groups for years of diagnosis. Test of Equality over Strata.



I. The Kaplan Meier curves for metastatic breast cancer patients with different groups for years of diagnosis. Test of Equality over Strata.

- A. The Kaplan Meier curves for metastatic breast cancer patients with years of diagnosis groups 1, 2, 3, and 4. Years 1988-2009 $\chi^2=105.843$, $P<.0001$.
- B. The Kaplan Meier curves for metastatic breast cancer patients with years of diagnosis groups 1 and 2. Years 1988-1998 $\chi^2=30.068$, $P=0.813$.
- C. The Kaplan Meier curves for metastatic breast cancer patients with years of diagnosis groups 2 and 3. Years 1998-2003 $\chi^2=74.564$, $P<.0001$.
- D. The Kaplan Meier curves for metastatic breast cancer patients with years of diagnosis groups 3 and 4. Years 2003-2009, $\chi^2=74.564$, $P<.0001$.
- E. The Kaplan Meier curves for metastatic breast cancer patients with years of diagnosis groups 1 and 4. Years 1988-2009 $\chi^2=105.843$, $P<.0001$.

II. The Kaplan Meier curves for metastatic breast cancer patients in different age groups (1-5), over different years of diagnosis.

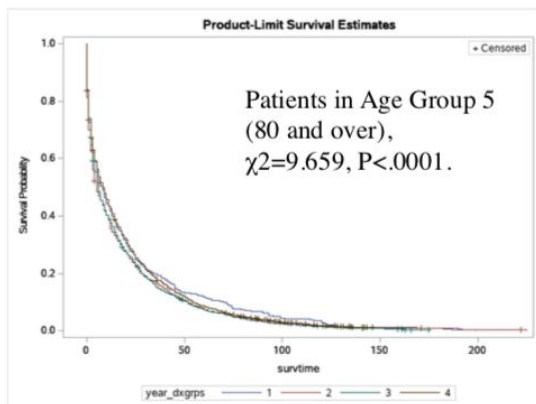
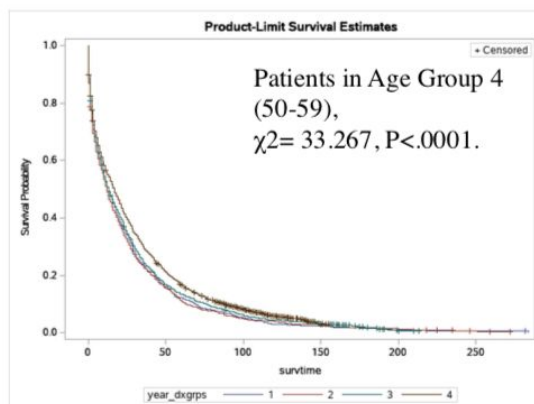
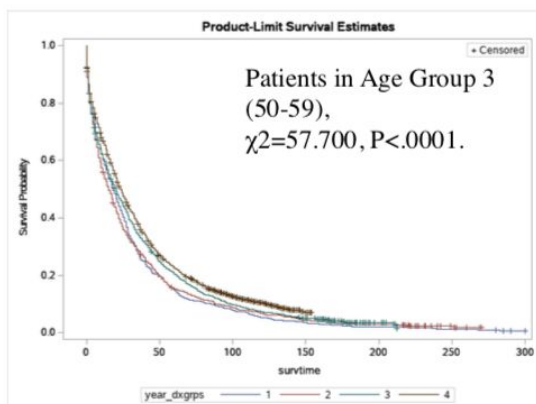
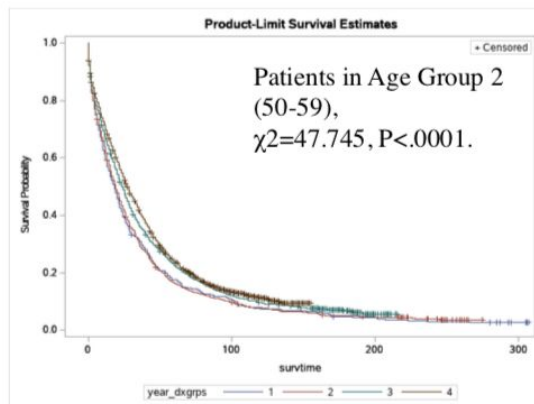
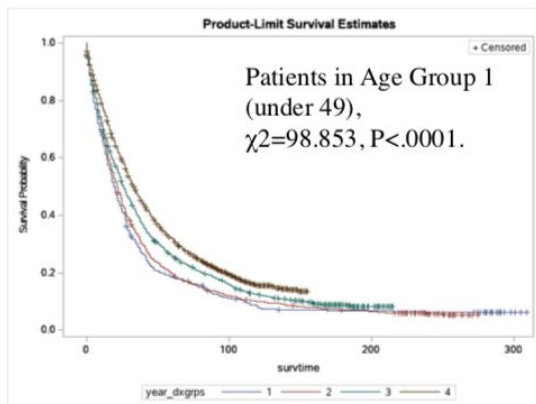


II. The Kaplan Meier curves for metastatic breast cancer patients in different age groups (1-5), over different years of diagnosis. Test of Equality over Strata.

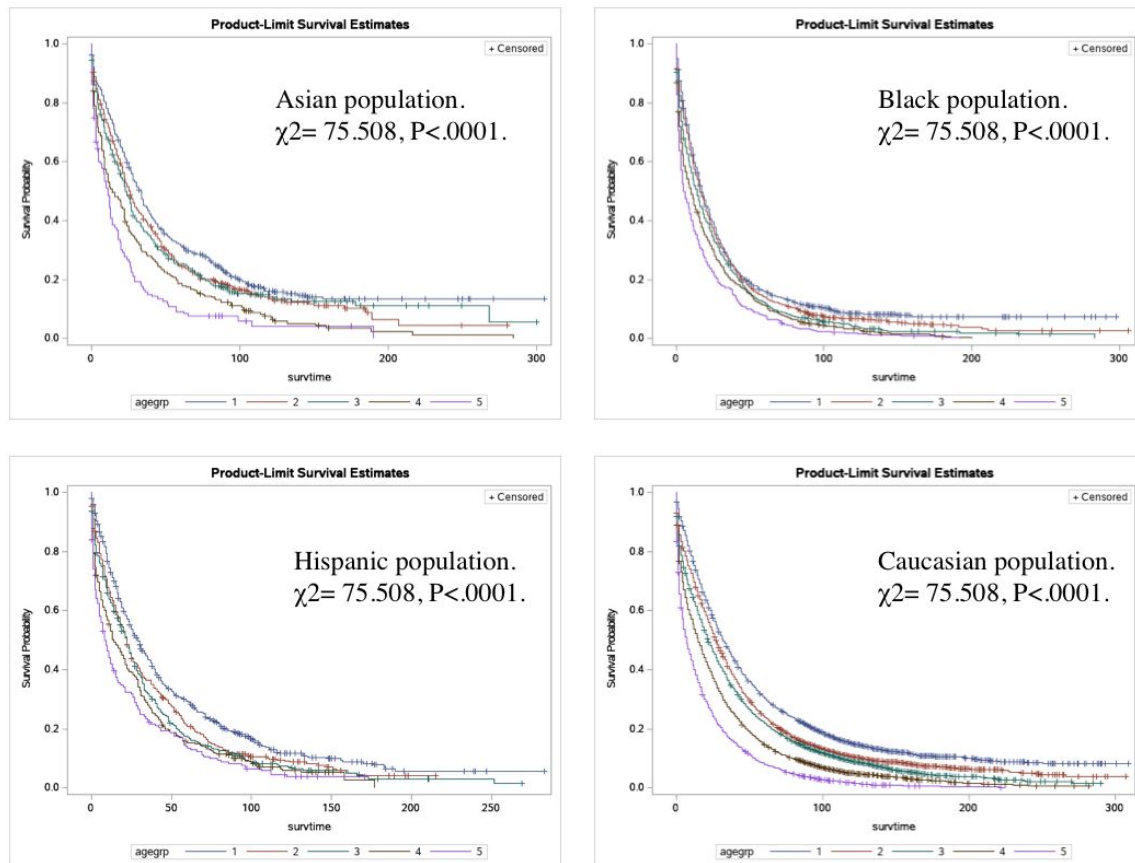
Age groups; group 1 (49 years and under), age group 2 (50-59), age group 3 (60-69), age group 4 (70-79), and age group 5 (80 and above).

- The Kaplan Meier curves for metastatic breast cancer patients diagnosed between years 1988-1993, for different age groups. $\chi^2 = 105.843, P < .0001$.
- The Kaplan Meier curves for metastatic breast cancer patients diagnosed between years 1993-1997. $\chi^2 = 213.327, P < .0001$.
- The Kaplan Meier curves for metastatic breast cancer patients diagnosed between years 1998-2003. $\chi^2 = 565.355, P < .0001$.
- The Kaplan Meier curves for metastatic breast cancer patients diagnosed between years 2003-2009. $\chi^2 = 1091.004, P < .0001$.

III. The Kaplan Meier curves for metastatic breast cancer patients in each age groups (1-5), over different years of diagnosis. Test of Equality over Strata.



IV. The Kaplan Meier curves for metastatic breast cancer patients in different age groups (1-5), for different races.



IV. The Kaplan Meier curves for metastatic breast cancer patients in different age groups (1-5), for different races. Test of Equality over Strata.

Races; Caucasian, Hispanic (Mexican, Puerto Rican, Cuban, Spanish/Hispanic/Latino, Dominican), Black, Asian, and Other.

- A. The Kaplan Meier curves for metastatic breast cancer patients diagnosed between years 1988-2009, for the Asian population. $\chi^2 = 75.508, P < .0001$.
- B. The Kaplan Meier curves for metastatic breast cancer patients diagnosed between years 1988-2009, for the Black population. $\chi^2 = 129.604, P < .0001$.
- C. The Kaplan Meier curves for metastatic breast cancer patients diagnosed between years 1988-2009, for the Hispanic population. $\chi^2 = 71.947, P < .0001$.
- D. The Kaplan Meier curves for metastatic breast cancer patients diagnosed between years 1988-2009, for the Caucasian population. $\chi^2 = 1750.815, P < .0001$.

Correlation Matrix

The CORR Procedure

5 Variables:	year_dxgrps	agegrp	num_race	num_marriage_stat	erpr
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Simple Statistics						
Variable	N	Mean	Std Dev	Sum	Minimum	Maximum
year_dxgrps	11820	3.47640	0.82818	41091	1.00000	4.00000
agegrp	11820	2.66827	1.30702	31539	1.00000	5.00000
num_race	11820	1.61371	1.00764	19074	1.00000	5.00000
num_marriage_stat	11820	1.47174	0.49922	17396	1.00000	2.00000
erpr	11820	1.81988	1.99995	21511	1.00000	9.00000

Pearson Correlation Coefficients, N = 11820 Prob > r under H0: Rho=0					
	year_dxgrps	agegrp	num_race	num_marriage_stat	erpr
year_dxgrps	1.00000	-0.04542 <.0001	0.05072 <.0001	-0.01993 0.0303	-0.05843 <.0001
agegrp	-0.04542 <.0001	1.00000	-0.15172 <.0001	-0.20738 <.0001	0.02178 0.0179
num_race	0.05072 <.0001	-0.15172 <.0001	1.00000	-0.06173 <.0001	0.02606 0.0046
num_marriage_stat	-0.01993 0.0303	-0.20738 <.0001	-0.06173 <.0001	1.00000	-0.02624 0.0043
erpr	-0.05843 <.0001	0.02178 0.0179	0.02606 0.0046	-0.02624 0.0043	1.00000

References

- (1) Surveillance, Epidemiology, and End Results (SEER) Program (www.seer.cancer.gov)
SEER*Stat Database: Incidence - SEER 9 Regs Research Data, Nov 2017 Sub (1973-2015)
<Katrina/Rita Population Adjustment> - Linked To County Attributes - Total U.S., 1969-2016

Counties, National Cancer Institute, DCCPS, Surveillance Research Program, released April 2018, based on the November 2017 submission.

(2) Smith, Ryan et al. "Survival Analysis Using Cox Proportional Hazards Modeling For Single And Multiple Event Time Data". Statistics and Data Analysis, SUGI28.
“<http://www2.sas.com/proceedings/sugi28/254-28.pdf>”.